

Optimality conditions: linear optimization

Reduced costs

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Reduced costs

In unconstrained optimization, the gradient characterizes the optimum because it reports the slope of the function.

In linear optimization, the gradient is constant.

We will use it to calculate the slope of the objective function along feasible directions.

Basis representation

$$\min_{x \in \mathbb{R}^n} f(x) = c^T x$$

subject to

$$Ax = b$$

$$x \geq 0.$$

$$A \in \mathbb{R}^{m \times n}$$

$$b \in \mathbb{R}^m$$

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Slope along the basic direction

$$\nabla f(x)^T d_j = c^T d_j = c_B^T (d_j)_B + c_N^T (d_j)_N = -c_B^T B^{-1} A_j + c_j$$

Reduced costs

Definition

$$\bar{c}_j = c_j - c_B^T B^{-1} A_j$$

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Non basic variables

Slope of the objective function along the corresponding basic direction.

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$$\bar{c}_j = c_j - c_B^T B^{-1} A_j = c_j - c_B^T e_j = c_j - c_j = 0$$

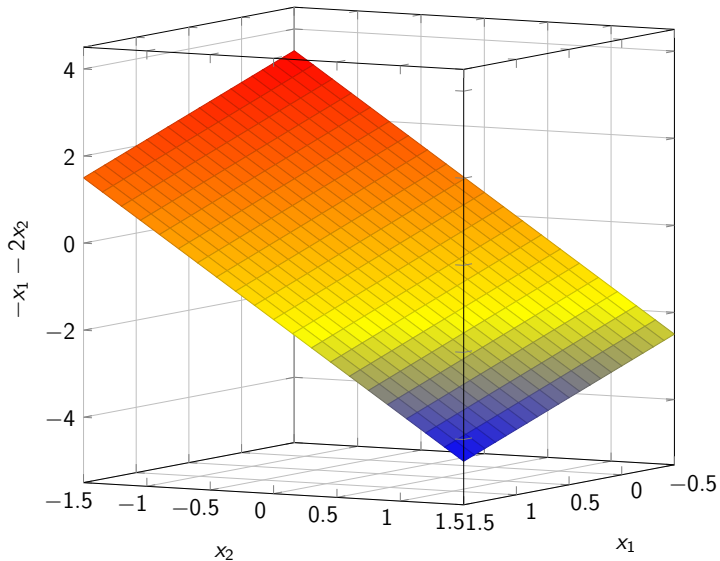
Matrix form

$$\bar{c} = c - A^T B^{-T} c_B.$$

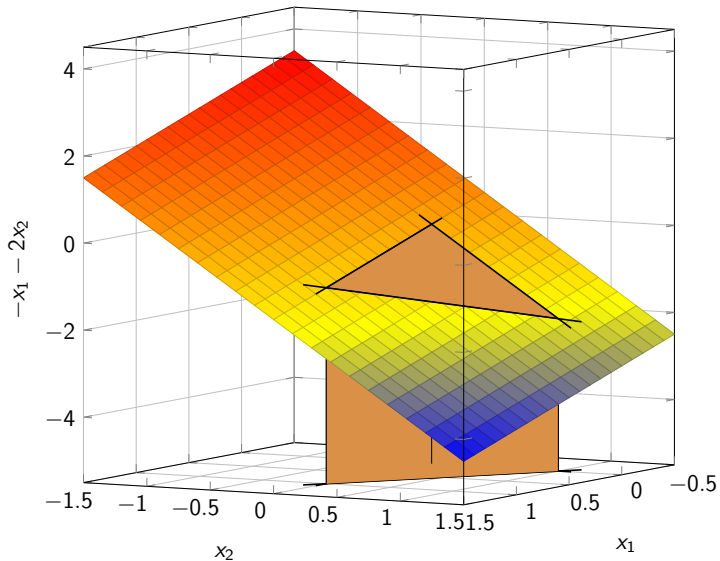
Sufficient optimality conditions

If $\bar{c} \geq 0$, then x^* is optimal.

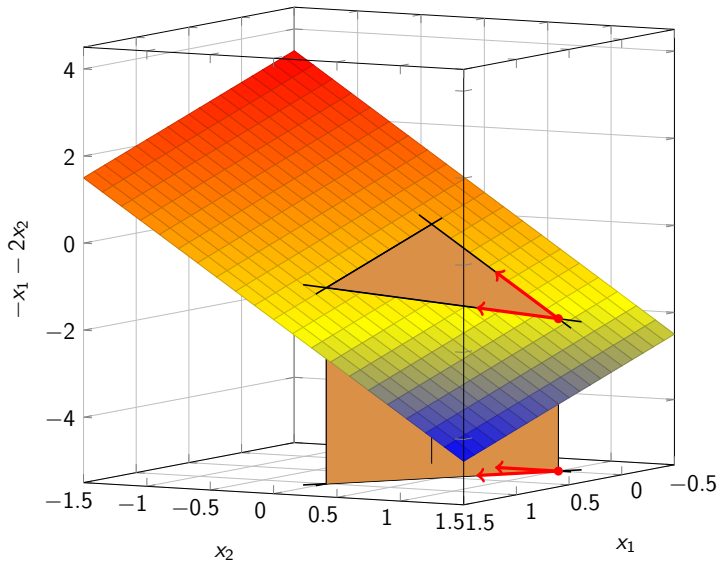
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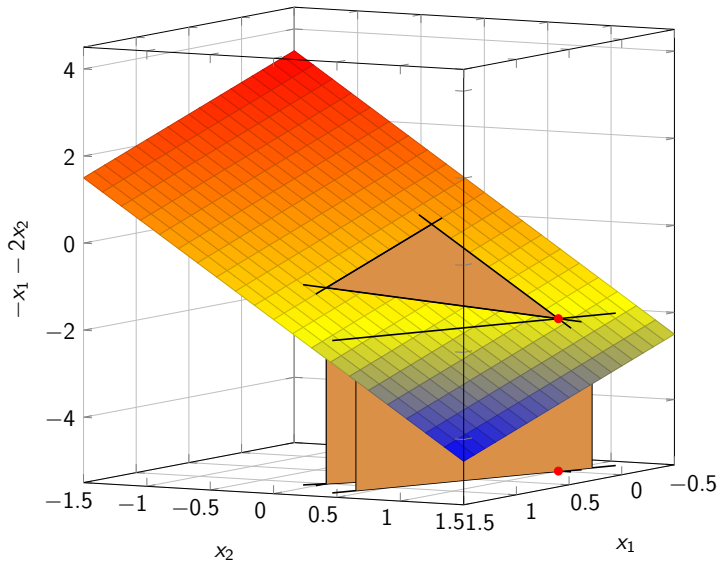
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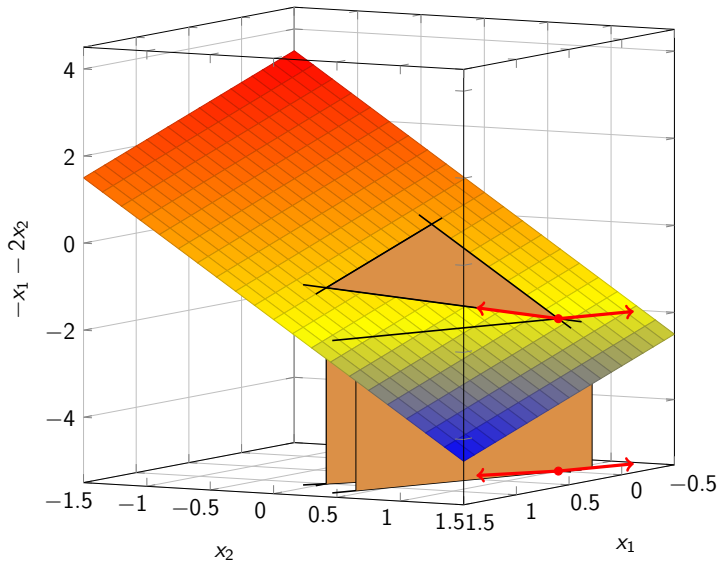
Necessary optimality conditions

- ▶ $\bar{c} \geq 0$ is not necessary at an optimal solution.
- ▶ Consider a slightly different example.

Necessary optimality conditions



Necessary optimality conditions



Necessary optimality conditions

- ▶ x^* is a **non degenerate** feasible basic solution.

Necessary optimality conditions

- ▶ x^* is a **non degenerate** feasible basic solution.
- ▶ If x^* is optimal, then

$$\bar{c} = c - A^T B^{-T} c_b \geq 0.$$

Necessary optimality conditions

Warning

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- ▶ If the basic solution is degenerate, it is possible that $c_j < 0$ for some j .
- ▶ It means that this is a descent direction.
- ▶ As x^* is optimal, d_j is infeasible.

Summary

The reduced costs represent the slope of the objective function along the basic directions

If they are non negative, it means that the solution is optimal

If the optimal solution is degenerate, it may happen that some reduced costs are negative.

The corresponding basic directions are therefore infeasible.